

ESTIMATION OF PARAMETRIC FUNCTIONS IN TWO-DIMENSIONAL SPACE IN SURVEY SAMPLING

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Summary: In this paper, the problem of estimation of parametric functions in two-dimensional plane is discussed. For a measurable design, a new estimator for estimating the variance of the Horvitz-Thompson estimator is suggested. The variance of the resultant estimator is discussed and compared with the Sen-Yates-Grundy estimator for a fixed sample design. An improved estimator of variance of the general linear regression estimator is also suggested. Techniques for estimating the ratio of two variances, the regression coefficient and the proportion of values within a specified region of a plane are also discussed. A simulation study has been carried out to support the ideas presented in the present investigation. The idea of obtaining different strategies is an extension of the pioneering methodology of Godambe from one-dimensional to two-dimensional space.

1. Introduction

Although for estimation of parametric functions we consider the extension of Godambe's methodology from one-dimensional to two-dimensional space, we

will quote his extensive work only through the recent reference of Thompson's (1997) book. Following Thompson (1997), let us consider the problem of estimation of population parameters which can be implicitly defined by a population equation of the form

$$\sum_{j=1}^N \phi_j (y_j, x_j, \theta) = 0 \quad (1.1)$$

where y_j and x_j are values of observed variates, y_j being real; the ϕ_j are known real-valued functions; and θ is a real-valued quantity defined by (1.1). Some special cases of (1.1) are as follows:

(a) the population mean μ_y defined by

$$\sum_{j=1}^N (y_j - \theta) = 0 \quad (1.2)$$

(b) the population total t_y defined by

$$\sum_{j=1}^N \left(y_j - \frac{t_y}{N} \right) = 0 \quad (1.3)$$

(c) the population ratio R of y to x defined by

$$\sum_{j=1}^N (y_j - Rx_j) = 0 \quad (1.4)$$

(d) the population c.d.f. evaluated at y defined as

$$\sum_{j=1}^N [I(y_j \leq y) - \theta] = 0 \quad (1.5)$$

where

$$I(y_j \leq y) = \begin{cases} 1 & \text{if } y_j \leq y \\ 0 & \text{otherwise} \end{cases}$$

(e) the population median defined as the least value of θ such that

$$\sum_{j=1}^N \left[I(y_j \leq \theta) - \frac{\theta}{2} \right] = 0 \quad (1.6)$$

When a population function (or parameter) is defined by (1.1), its estimator can be defined as a solution of the sample estimating equation

$$\sum_{j \in s} \phi_j(y_j, x_j, \theta) / \pi_j = 0 \quad (1.7)$$

where π_j denotes the probability of including the j th unit in the sample.

An estimator of the population mean can easily be obtained by solving

$\sum_{j \in s} (y_j - \theta) / \pi_j = 0$, that is,

$$\bar{y}_s = \left(\sum_{j \in s} \frac{y_j}{\pi_j} \right) / \left(\sum_{j \in s} \frac{1}{\pi_j} \right) \quad (1.8)$$

An estimator of the population total can be obtained by solving

$\sum_{j \in s} (y_j - \theta/N) / \pi_j = 0$, that is,

$$\hat{y}_s = \left(N \sum_{j \in s} \frac{y_j}{\pi_j} \right) / \left(\sum_{j \in s} \frac{1}{\pi_j} \right). \quad (1.9)$$

Similarly estimators of the ratio R of y to x , median of y and c.d.f. of y can be obtained by following Thompson (1997).

We have seen that all these estimators are based on first order inclusion probabilities. We also know that the estimators of variance depend upon the second order inclusion probabilities. A new estimator of variance has been suggested by casting the problem of estimating the variance in an estimating equation framework. A simulation study is used to show improvements in the coverage of the associated confidence intervals.

2. Proposed strategies

We consider the general parameter of interest, defined as

$$\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \Psi [y_i, y_j, x_i, x_j, \Theta] = 0. \quad (2.1)$$

The indices i and j may or may not be equal. If $i = j$, the proposed strategy (2.1) reduces to the one discussed in Section 1. An unbiased estimator of (2.1) for a measurable design will be given by

$$\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\Psi [y_i, y_j, x_i, x_j, \Theta]}{\pi_{ij}} = 0. \quad (2.2)$$

where $\pi_{ij} = \Pr(i \text{ \& } j \in s)$ denotes the probability of including i th and j th units in the sample and is assumed to be strictly positive and known.

3. A new estimator of variance

Let us now consider the following function to be estimated, given by

$$\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i=1}^N \left[\Omega_{ij} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 - \frac{2V}{N(N-1)} \right] = 0 \quad (3.1)$$

where $\Omega_{ij} = \pi_i \pi_j - \pi_{ij}$ and $V = \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i=1}^N \Omega_{ij} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2$ denotes the parameter, that is, variance to be estimated and $2/N(N-1)$ denotes an adjustment factor. Using (2.2), an unbiased estimator of (3.1) is given by

$$\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \left[\Omega_{ij} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 - \frac{2V}{N(N-1)} \right] / \pi_{ij} = 0 \quad (3.2)$$

which leads to the following new estimator of variance of Horvitz and Thompson (1952) estimator in Sen (1953) and Yates and Grundy (1953)

(referred to as Sen-Yates-Grundy in this paper) form given by

$$\hat{V}_2 = \frac{\frac{N(N-1)}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2}{\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}}} = \hat{V}_1 \left(\frac{N(N-1)}{\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}}} \right) \quad (3.3)$$

where $\hat{V}_1 = \frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2$ denotes the usual Sen-Yates-Grundy estimator of variance for the fixed sample and a measurable design.

In general, we have

$$\begin{aligned} Var(\hat{V}_2) &= V(\hat{V}_1) + \left\{ \frac{V}{N(N-1)} \right\}^2 V \left(\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}} \right) \\ &\quad - 2 \left\{ \frac{V}{N(N-1)} \right\} Cov \left(\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}}, \hat{V}_1 \right) \end{aligned} \quad (3.4)$$

$$\begin{aligned} Var(\hat{V}_1) &= \frac{1}{4} \sum_{i=1}^N \sum_{j=1}^N \sum_{k=1}^N \sum_{l=1}^N (\pi_{ijkl} - \pi_{ij}\pi_{kl}) \left(\frac{\pi_i\pi_j - \pi_{ij}}{\pi_{ij}} \right) \\ &\quad \cdot \left(\frac{\pi_k\pi_l - \pi_{kl}}{\pi_{kl}} \right) \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 \left(\frac{y_k}{\pi_k} - \frac{y_l}{\pi_l} \right)^2 \end{aligned} \quad (3.5)$$

$$\begin{aligned} Var \left(\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}} \right) &= \sum_{i=1}^N \sum_{j \neq i=1}^N \left[\frac{1}{\pi_{ij}} + \sum_{k \neq j \neq i=1}^N \sum_{l \neq j \neq k \neq i=1}^N \frac{\pi_{ijkl}}{\pi_{ij}\pi_{kl}} \right] \\ &\quad - \{N(N-1)\}^2 \end{aligned} \quad (3.6)$$

$$\begin{aligned} Cov \left[\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}}, \hat{V}_1 \right] &= \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i=1}^N (\pi_i\pi_j - \pi_{ij}) \\ &\quad \cdot \left[\frac{1}{\pi_{ij}} + \sum_{k \neq j \neq i=1}^N \sum_{l \neq k \neq j \neq i=1}^N \frac{\pi_{ijkl}}{\pi_{kl}} - \frac{N(N-1)}{2} \right] \\ &\quad \cdot \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 \end{aligned} \quad (3.7)$$

The equations (3.5), (3.6) and (3.7) depend upon the third and fourth order inclusion probabilities, recently used by Shah and Patel (1996). Evidently the proposed estimator $\hat{V}_2$ will be more efficient than $\hat{V}_1$ if

$$V < N(N-1) \frac{Cov \left(\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}}, \hat{V}_1 \right)}{Var \left(\sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}} \right)}. \quad (3.8)$$

This is analogous to the condition under which ratio-type estimators are more efficient than simple estimators.

Special cases:

(a) Under SI design (Särndal *et al.*, 1992, p. 61), we have $\pi_i = \pi_j = n/N$ and $\pi_{ij} = n(n-1)/N(N-1)$. Under such situations, both the estimators of variance $\hat{V}_1$ and $\hat{V}_2$ reduce to $\frac{N^2(1-f)}{n} s_y^2$ with $f = n/N$.

(b) **Midzuno system of sampling:** Following Asok and Sukhatme (1978), for Midzuno (1952) scheme with revised selection probabilities, the first and second order inclusion probabilities are given by $\pi_i = nP_i$ and $\pi_{ij} = \frac{n(n-1)}{(N-2)} \left\{ P_i + P_j - \frac{1}{N-1} \right\}$ respectively. Thus the proposed estimator reduces to

$$\hat{V}_2 = \frac{\frac{1}{2n(N-2)} \sum_{i=1}^n \sum_{j \neq i=1}^n \left\{ n(N-2)P_iP_j \left(P_i + P_j - \frac{1}{N-1} \right)^{-1} - (n-1) \right\} \left(\frac{y_i}{P_i} - \frac{y_j}{P_j} \right)^2}{\{N(N-1)\}^{-1} \sum_{i=1}^n \sum_{j \neq i=1}^n \left(P_i + P_j - \frac{1}{N-1} \right)^{-1}} \quad (3.9)$$

and $\hat{V}_1$ is given by

$$\hat{V}_1 = \frac{1}{2n(N-2)} \sum_{i=1}^n \sum_{j \neq i=1}^n \left\{ n(N-2)P_i P_j \left(P_i + P_j - \frac{1}{N-1} \right)^{-1} - (n-1) \right\} \cdot \left(\frac{y_i}{P_i} - \frac{y_j}{P_j} \right)^2 \quad (3.10)$$

In order to study how $\hat{V}_2$ may fare with respect to $\hat{V}_1$ in actual practice when the values of n and N are moderately large, we consider a simulation study to produce appropriate confidence intervals (CI) for the population total under the Midzuno (1952) scheme of sampling. To construct a CI for Y , with $\hat{Y}_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i}$ as a point estimator and $\hat{V}_1$ (or $\hat{V}_2$) as the variance estimator for it, we regard $(\hat{Y}_{HT} - Y) / \sqrt{\hat{V}_j}$, $j = 1, 2$, as a variable distributed approximately as a standardised normal variate Z for large n . Thus for a given Z in $(0, 1)$, say z_0 , for the $100\alpha\%$ point on the right tail area of the normal distribution, $\hat{Y}_{HT} \mp z_{\alpha/2} \sqrt{\hat{V}_j}$, $j = 1, 2$, gives $100(1-\alpha)\%$ CI for Y , with α in $(0, 1)$. Following Chaudhuri and Roy (1997) we take $N = 160$ and different sample sizes $n = 30, 50$ and 75 . To keep x_i positive, we take $x_i = 10 + \theta_i$ generating θ_i as random samples from the negative exponential density $f(x) = \lambda \exp(-\lambda x)$, $\lambda > 0$, $x > 0$, taking different values of λ . We also considered $f_i = x_i^{c_0}$, with $c_0 = 1.1$ only. Drawing $z_i (i = 1, 2, \dots, N)$ as random samples from $N(0, 1)$, taking $\varepsilon_i = \sigma z_i \sqrt{f_i}$ and choosing (μ, σ) differently we generated Y 's with the help of subroutine VNORM available in Bratley *et al.* (1983). To draw samples from U we apply Midzuno's scheme of selection drawing first a unit from U with a probability proportional to size measures x_i taken as the highest integer not exceeding $8.2 + 0.65x_i^{0.78}$, the remaining $(n-1)$ units are chosen at random without replacement

from the $(N - 1)$ units left in U after the first draw. We take the number of replicates as 25,000. The 95% confidence intervals by both estimators of variance were obtained following Singh *et al.* (1999). Results obtained are presented in Table 1. It is shown that the proposed estimator remains better than the usual Sen-Yates-Grundy estimator under Midzuno's scheme of sampling.

- (c) **PPSWR sampling:** Here we have $\pi_i = nP_i$ and $\pi_{ij} = n(n - 1)P_iP_j$. The proposed estimators of variance reduce to

$$\hat{V}_2 = \frac{\frac{1}{2n} \sum_{i=1}^n \sum_{j \neq i=1}^n \left(\frac{y_i}{P_i} - \frac{y_j}{P_j} \right)^2}{\{N(N - 1)\}^{-1} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{P_iP_j}} \quad (3.11)$$

and

$$\hat{V}_1 = \frac{1}{2n^2(n-1)} \sum_{i=1}^n \sum_{j \neq i=1}^n \left(\frac{y_i}{P_i} - \frac{y_j}{P_j} \right)^2 = \frac{1}{n(n-1)} \sum_{i=1}^n \left(\frac{y_i}{P_i} - \hat{Y}_{HH} \right)^2 \quad (3.12)$$

where $\hat{Y}_{HH} = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{P_i}$ denotes the Hansen and Horwitz (1943) estimator of the population total. In case of PPSWR sampling, we considered several natural populations. In each population, two variables, Y and X (say) have been chosen. The variable X is treated as an auxiliary variable while Y is treated as the study variable. The auxiliary variable X is used at the selection stage. We selected 25,000 samples of different sizes by using the Cumulative-total method. We used a subroutine INTEGER from Bratley *et al.* (1983) to select the random numbers required to use the Cumulative-total method. The 95% confidence intervals for the estimator of the total were obtained by counting the number of times the true value of

the population total for the study variable (Y) falls in the intervals

$$\hat{Y}_{HH} \mp 1.96 \sqrt{\hat{V}_j}, \quad j = 1, 2 \quad (3.13)$$

The results obtained are presented in Table 2.

4. Improved higher order calibrated estimators of variance

Following Godambe (1995), let $U_{jk}(\Theta)$ denote the robust optimal estimating function for the j th and k th units of the survey population. Then the induced parameter solves the equation

$$\sum_{j=1}^N \sum_{k=1}^N U_{jk}(\Theta) = 0. \quad (4.1)$$

Singh *et al.* (1998, 1999) have considered the difference ratio and regression type estimators for estimating the variance of the general regression estimator in the presence of known variance of the auxiliary character. If $f_{jk}(y_j, y_k)$ and $\alpha_{jk}(\Theta)$ are the specified functions of the indicated variables, then an optimal estimator of

$$\sum_{j=1}^N \sum_{k=1}^N f_{jk}(y_j, y_k) \quad (4.2)$$

is given by

$$\sum_{j \in s} \sum_{k \in s} \frac{f_{jk}(y_j, y_k) - \alpha_{jk}(\Theta)}{\pi_{jk}} + \sum_{j=1}^N \sum_{k=1}^N \alpha_{jk}(\Theta). \quad (4.3)$$

For example, a difference type estimator of variance is given by

$$\begin{aligned}
& \sum_{j \in s} \sum_{k \in s} \frac{(\pi_j \pi_k - \pi_{jk}) \left[\left(\frac{y_j}{\pi_j} - \frac{y_k}{\pi_k} \right)^2 - \left(\frac{x_j}{\pi_j} - \frac{x_k}{\pi_k} \right)^2 \right]}{\pi_{jk}} \\
& + \sum_{j=1}^N \sum_{k=1}^N (\pi_j \pi_k - \pi_{jk}) \left(\frac{x_j}{\pi_j} - \frac{x_k}{\pi_k} \right)^2
\end{aligned} \quad (4.4)$$

The estimator (4.4) is a member of the higher order calibration approach suggested by Singh *et al.* (1998, 1999). A new estimator of variance of the regression estimator has been suggested as

$$\hat{V}_2 \left(\hat{Y}_{reg} \right) = \frac{\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \Omega_{ij}^* (w_i e_i - w_j e_j)^2}{\{N(N-1)\}^{-1} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{1}{\pi_{ij}}} \quad (4.5)$$

where

$$\begin{aligned}
\Omega_{ij}^* &= \frac{\Omega_{ij}}{\pi_{ij}} + \frac{\frac{\Omega_{ij}}{\pi_{ij}} Q_{ij} \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2}{\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\Omega_{ij}}{\pi_{ij}} Q_{ij} \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^4} \\
& \left[V_1 \left(\hat{X}_{HT} \right) - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2 \right]
\end{aligned} \quad (4.6)$$

and

$$w_i = \frac{1}{\pi_i} + \frac{q_i x_i / \pi_i}{\sum_{i=1}^n q_i x_i^2 / \pi_i} \left(X - \hat{X}_{HT} \right) \quad (4.7)$$

where q_i and Q_{ij} are suitably chosen constants, which are the higher and lower order calibrated weights as reported by Singh *et al.* (1998, 1999).

$\hat{X}_{HT} = \sum_{i=1}^n \frac{x_i}{\pi_i}$ is an unbiased estimator of the known total $X = \sum_{i=1}^N X_i$

of the auxiliary character and $V_1 \left(\hat{X}_{HT} \right)$ denotes the known variance of the

auxiliary character. Under SI design, the proposed estimators $\hat{V}_2(\hat{Y}_{greg})$ defined in (4.5) and higher order calibrated estimator defined as

$$\hat{V}_1(\hat{Y}_{greg}) = \frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \Omega_{ij}^* (w_i e_i - w_j e_j)^2 \quad (4.8)$$

are the same. Both estimators are different under PPSWR sampling, Midzuno's method of sampling and several other sampling schemes available in Brewer and Hanif (1983). If $q_i = 1/x_i$ and $Q_{ij} = \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^{-2}$, then the estimators $\hat{V}_1(\hat{Y}_{greg})$ and $\hat{V}_2(\hat{Y}_{greg})$ reduce to $\hat{V}_1(\hat{Y}_{ratio})$ and $\hat{V}_2(\hat{Y}_{ratio})$ respectively, where the estimators $\hat{Y}_{ratio}$ and $\hat{Y}_{greg}$ denote the usual ratio and regression estimators of the population total as defined by Singh *et al.* (1999) for the PPSWR sampling design. A condition of efficiency analogous to (3.8) such that the estimator $\hat{V}_2(\hat{Y}_{greg})$ is more efficient than $\hat{V}_1(\hat{Y}_{greg})$ is straightforward. We compare these estimators through simulation under the PPSWR design only. In the case of PPSWR sampling, we again consider several natural populations. In each population, three variables, Y , X and Z (say) have been chosen. The variables X and Z are treated as auxiliary variables while Y is treated as the study variable. The auxiliary variable Z is used at the selection stage and X has been used at the estimation stage. We selected 25,000 samples of different sizes by using the Cumulative-total method. We used a subroutine INTEGER from Bratley *et al.* (1983) to select the random numbers required to use the Cumulative-total method. The 95% confidence intervals for the estimator of the total were obtained by counting the number of times the true value of the population total for study variable (Y) falls in the intervals

$$\hat{Y}_{ratio} \mp t_{n-3} \sqrt{\hat{V}_j(\hat{Y}_{ratio})} \quad j = 1, 2 \quad (4.9)$$

and

$$\hat{Y}_{greg} \mp t_{n-3} \sqrt{\hat{V}_j \left(\hat{Y}_{greg} \right)} \quad j = 1, 2 \quad (4.10)$$

The results obtained are presented in Table 3. It is observed that the proposed estimator performs better than the higher order calibration estimator proposed by Singh *et al.* (1999) under PPSWR sampling. In other words, the coverage by the proposed estimators converges to the real coverage more rapidly than Singh *et al.* (1999) strategies.

Remark: In the absence of known variance of the auxiliary character, an estimator of variance of the ratio estimator reported by Rao (1997),

$$s^2 \left(\hat{Y}_r \right) = \frac{1}{2} \sum_{i \in s} \sum_{j \neq i \in s} \frac{\pi_i \pi_j - \pi_{ij}}{\pi_{ij}} \left(\frac{e_i}{\pi_i} g_i - \frac{e_j}{\pi_j} g_j \right)^2 \quad (4.11)$$

where $g_i = X/\hat{X}_{HT}$ and $e_i = y_i - \left(\hat{Y}_{HT}/\hat{X}_{HT} \right) x_i$ is also a member of the two-dimensional strategy proposed in the present investigation. According to the proposed strategy, an improved estimator of the variance of the ratio estimator is given by

$$s_I^2 \left(\hat{Y}_r \right) = \frac{\frac{1}{2} \sum_{i \in s} \sum_{j \neq i \in s} \frac{\pi_i \pi_j - \pi_{ij}}{\pi_{ij}} \left(\frac{e_i}{\pi_i} g_i - \frac{e_j}{\pi_j} g_j \right)^2}{\{N(N-1)\}^{-1} \sum_{i \in s} \sum_{j \neq i \in s} \frac{1}{\pi_{ij}}} \quad (4.12)$$

5. Some applications of the proposed strategy

We list below some more applications of the proposed strategy in actual surveys:

5.1 Estimation of variance ratio

Let us now consider the problem of estimation of the ratio of variance of the study character to that of the auxiliary character as

$$\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i=1}^N \Omega_{ij} \left[\left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 - V_R \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2 \right] = 0 \quad (5.1)$$

Obviously for a measurable design, an estimator of (5.1) is given by

$$\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left[\left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 - V_R \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2 \right] = 0 \quad (5.2)$$

Thus an estimator of the variance ratio is given by

$$\hat{V}_R = \frac{\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2}{\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2} \quad (5.3)$$

Under SI design, the estimator $\hat{V}_R$ reduces to the estimator studied by Isaki (1983) and Garcia and Cebrian (1996).

5.2 Estimation of regression coefficient

Let us consider the problem of estimation of the regression coefficient as

$$\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i=1}^N \Omega_{ij} \left[\left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right) \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right) - \beta \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2 \right] = 0 \quad (5.4)$$

For a measurable design, an unbiased estimator of (5.4) is given by

$$\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left[\left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right) \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right) - \beta \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2 \right] = 0 \quad (5.5)$$

Thus an estimator of the regression coefficient is given by

$$\hat{\beta} = \frac{\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right) \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)}{\frac{1}{2} \sum_{i=1}^n \sum_{j \neq i=1}^n \frac{\Omega_{ij}}{\pi_{ij}} \left(\frac{x_i}{\pi_i} - \frac{x_j}{\pi_j} \right)^2} \quad (5.6)$$

Under SI design, the estimator $\hat{\beta}$ reduces to the estimator studied by Singh and Singh (1988).

5.3 Estimation of P_{11}

The problem of estimation of the median M_y of the study character in the presence of the known median M_x of the auxiliary character has been considered by Kuk and Mak (1989) by defining a matrix of proportions (P_{ij}) as

	$Y \leq M_y$	$Y > M_y$	Total
$X \leq M_x$	P_{11}	P_{21}	P_1
$X > M_x$	P_{12}	P_{22}	P_2
Total	P_1	P_2	1

They have considered three estimators $\hat{M}_R$, $\hat{M}_s$ and $\hat{M}_p$ under the names of ratio, stratified and position estimators of the median of y , respectively, in the presence of the known median of the variable, x . It is interesting to note that the variances of all three estimators are a function of P_{11} for which $X \leq M_x$ and $Y \leq M_y$ as defined above. The value of P_{11} can be written as

$$\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N [I(y_i \leq M_y \cap x_j \leq M_x) - P_{11}] = 0 \quad (5.7)$$

where

$$I(y_i \leq M_y \cap x_j \leq M_x) = \begin{cases} 1 & \text{if } y_i \leq M_y \cap x_j \leq M_x \\ 0 & \text{otherwise} \end{cases} \quad (5.8)$$

An unbiased estimator for (5.7) is given by

$$\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{[I(y_i \leq M_y \cap x_j \leq M_x) - P_{11}]}{\pi_{ij}} = 0 \quad (5.9)$$

or equivalently

$$\hat{P}_{11} = \frac{\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n [I(y_i \leq M_y \cap x_j \leq M_x) / \pi_{ij}]}{\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{1}{\pi_{ij}}} \quad (5.10)$$

Thus the proposed strategy may be applied while estimating the variances of the estimators proposed by Kuk and Mak (1989).

5.4 Two-way post-stratification

Let y and x be the two characters used for post-stratifying the given data set.

Let us define

$$y_i = \begin{cases} 1 & \text{if } a \leq y \leq b \\ 0 & \text{otherwise} \end{cases} \quad (5.11)$$

and

$$x_i = \begin{cases} 1 & \text{if } c \leq y \leq d \\ 0 & \text{otherwise} \end{cases} \quad (5.12)$$

where a, b, c and d are predefined characters or limits used for defining the stratum boundaries. For example, the strata defined by the equation

$$\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N [I(a \leq y_i \leq b, c \leq x_j \leq d) - W] = 0 \quad (5.13)$$

where

$$I(a \leq y_i \leq b, c \leq x_j \leq d) = \begin{cases} 1 & \text{if } a \leq y_i \leq b, c \leq x_j \leq d \\ 0 & \text{otherwise} \end{cases} \quad (5.14)$$

contains the number in the sample with $a \leq y_i \leq b$ and $c \leq x_j \leq d$. In case of post stratification, the sample size n is a random variable, therefore, an estimator of (5.13) is given by

$$\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n [I(a \leq y_i \leq b, c \leq x_j \leq d) - W] / \pi_{ij} = 0 \quad (5.15)$$

or equivalently

$$\hat{W} = \frac{\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n [I(a \leq y_i \leq b, c \leq x_j \leq d)] / \pi_{ij}}{\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{1}{\pi_{ij}}} \quad (5.16)$$

Conclusion

From the above discussion, we conclude that the proposed strategy is quite useful while estimating parameters dependent on second order inclusion probabilities in survey sampling. The proposed strategy leads to improved estimators of the variance of the regression, ratio and the usual estimator of the population total in survey sampling.

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Table 1. Comparison of $\hat{V}_1$ with $\hat{V}_2$ using Midzuno's scheme of sampling over 25,000 samples.

n	λ	μ	σ	CI ₁	CI ₂
30	0.6	50	10	0.915	0.952
			15	0.896	0.950
			20	0.886	0.942
		100	10	0.922	0.952
			15	0.932	0.951
			20	0.947	0.941
	0.4	50	10	0.914	0.958
			15	0.892	0.951
			20	0.886	0.941
		100	10	0.923	0.946
			15	0.934	0.949
			20	0.941	0.950
50	0.6	50	10	0.935	0.945
			15	0.922	0.946
			20	0.932	0.942
		100	10	0.942	0.947
			15	0.948	0.950
			20	0.942	0.952
	0.4	50	10	0.938	0.947
			15	0.923	0.943
			20	0.922	0.943
		100	10	0.947	0.948
			15	0.944	0.950
			20	0.942	0.951
100	0.6	50	10	0.942	0.951
			15	0.945	0.949
			20	0.943	0.948
		100	10	0.948	0.950
			15	0.947	0.951
			20	0.947	0.949
	0.4	50	10	0.946	0.948
			15	0.948	0.949
			20	0.946	0.947
		100	10	0.951	0.949
			15	0.952	0.950
			20	0.949	0.950

Table 2. Comparison of $\hat{V}_1$ with $\hat{V}_2$ using PPSWR over 25,000 samples.

Source	Variables	Correlations	N	n	CI ₁	CI ₂
Gunst and Mason (1980) p. 358	y: GNP per capita, 1957, US \$	$\rho_{xy} = 0.6458$	49	10	0.896	0.902
	x: Students enrolled in Higher education per 10 ⁵ population			15	0.923	0.946
				20	0.932	0.941
				30	0.946	0.950
Gunst and Mason (1980) p. 367	y: Height of black female applicants with the police department (cm)	$\rho_{xy} = 0.4949$	33	6	0.886	0.902
	x: Foot length (cm)			8	0.893	0.932
				10	0.921	0.942
				15	0.936	0.951

Table 3. Comparison of $\hat{V}_1(\hat{Y}_{ratio})$ with $\hat{V}_2(\hat{Y}_{ratio})$ and $\hat{V}_1(\hat{Y}_{greg})$ with $\hat{V}_2(\hat{Y}_{greg})$ using PPSWR over 25,000 samples.

Source	Variables	Correlations	N	n	Ratio estimator		Regression estimator	
					CI ₂	CI ₁	CI ₂	CI ₁
Gunst and Mason (1980) p. 358	y: GNP per capita 1957, US \$	$\rho_{xy} = 0.6458$	49	10	0.944	0.946	0.942	0.944
	x: Students enrolled in Higher education per 10 ⁵ population	$\rho_{xz} = 0.4173$		15	0.937	0.941	0.940	0.947
				20	0.942	0.950	0.949	0.950
	z: age Literate of population aged 15 and over.	$\rho_{yz} = 0.6042$		30	0.949	0.952	0.948	0.949
Gunst and Mason (1980) p. 367	y: Height of black female applicants with the police department (cm)	$\rho_{xy} = 0.4949$	33	6	0.892	0.899	0.922	0.936
		$\rho_{xz} = 0.3680$		8	0.894	0.921	0.935	0.945
	x: Foot length (cm)			10	0.892	0.926	0.942	0.950
	z: Sitting height (cm)	$\rho_{yz} = 0.6848$		15	0.924	0.937	0.942	0.946